LAB EXPERIMENT RECORD

14. Dataset: Survey Results

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
Name	Swetha S
Register Number	192424049
Date	08/10/2026
Day	thursday

AIM

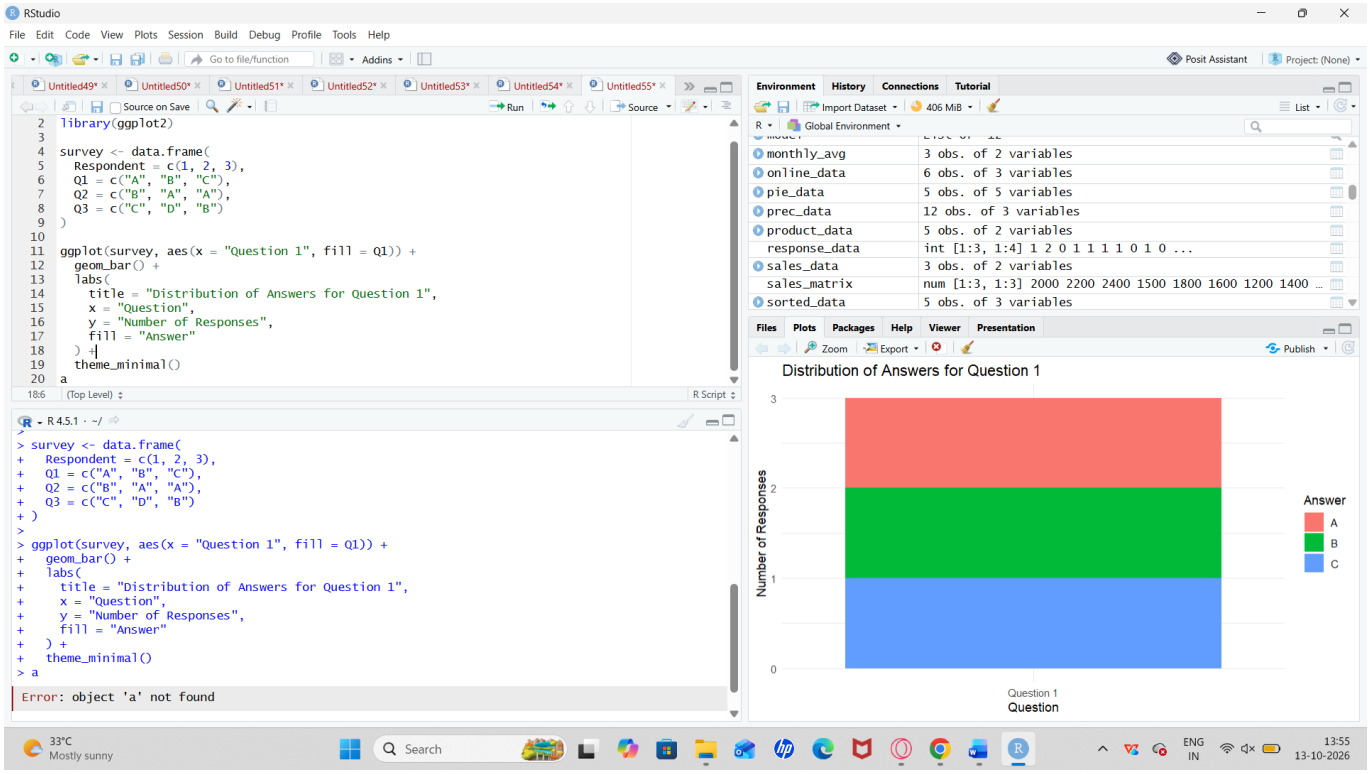
To analyze and visualize survey results using R programming by creating a stacked bar chart, radar chart, and data table, and to prepare an interactive Tableau dashboard for exploring survey responses across three questions.

1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

```
library(ggplot2)

survey <- data.frame(
  Respondent = c(1, 2, 3),
  Q1 = c("A", "B", "C"),
  Q2 = c("B", "A", "A"),
  Q3 = c("C", "D", "B")
)

ggplot(survey, aes(x = "Question 1", fill = Q1)) +
  geom_bar() +
  labs(
    title = "Distribution of Answers for Question 1",
    x = "Question",
    y = "Number of Responses",
    fill = "Answer"
  ) +
  theme_minimal()
a
```



2. Generate a radar chart to visualize the overall pattern of responses across all three questions..

```
library(ggplot2)

# Survey response frequencies
responses <- data.frame(
  Question = rep(c("Q1", "Q2", "Q3"), each = 4),
  Answer = rep(c("A", "B", "C", "D"), 3),
  Count = c(1, 1, 1, 0,
            2, 1, 0, 0,
            0, 1, 1, 1)
)

# Convert to radar chart coordinates
responses$Angle <- 2 * pi *
  (match(responses$Question, c("Q1", "Q2", "Q3")) - 1) / 3

responses$Radius <- responses$Count

ggplot(responses,
       aes(x = Angle, y = Radius,
           color = Answer, group = Answer)) +
  geom_polygon(fill = NA, linewidth = 1) +
  geom_point(size = 3) +
  scale_x_continuous(
    breaks = 0:(2) * 2 * pi / 3,
    labels = c("Q1", "Q2", "Q3"),
    limits = c(-0.2, 2 * pi + 0.2)
  ) +
  coord_polar() +
  labs(
    title = "Survey Response Pattern",
    x = "Question",
    y = "Response Frequency",
    color = "Answer"
  ) +
  theme_minimal()
```

RStudio

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```
1
2
3  library(ggplot2)
4
5  # Survey response frequencies
6  responses <- data.frame(
7    Question = rep(c("Q1", "Q2", "Q3"), each = 4),
8    Answer = rep(c("A", "B", "C", "D"), 3),
9    Count = c(1, 1, 1, 0,
10             2, 1, 0, 0,
11             0, 1, 1, 1)
12 )
13
14 # Convert to radar chart coordinates
15 responses$Angle <- 2 * pi *
16   (match(responses$Question, c("Q1", "Q2", "Q3")) - 1) / 3
17
18 responses$Radius <- responses$Count
19
```

13:1 (Top Level) R Script

R 4.5.1 · ~/

```
>
> ggplot(responses,
+        aes(x = Angle, y = Radius,
+            color = Answer, group = Answer)) +
+   geom_polygon(fill = NA, linewidth = 1) +
+   geom_point(size = 3) +
+   scale_x_continuous(
+     breaks = 0:(2) * 2 * pi / 3,
+     labels = c("Q1", "Q2", "Q3"),
+     limits = c(-0.2, 2 * pi + 0.2)
+   ) +
+   coord_polar() +
+   labs(
+     title = "Survey Response Pattern",
+     x = "Question",
+     y = "Response Frequency",
+     color = "Answer"
+   ) +
+   theme_minimal()
>
```

Environment History Connections Tutorial

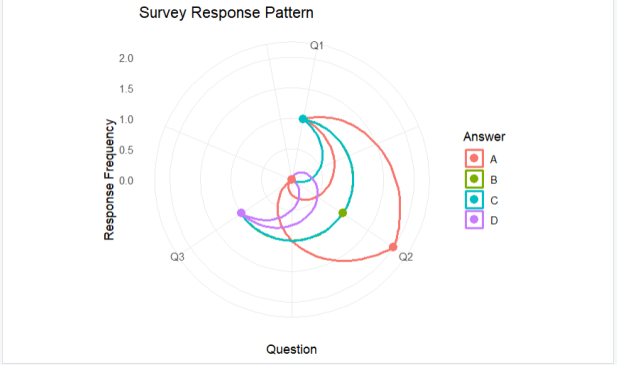
Import Dataset | 366 MiB | List

R | Global Environment

pie_data	5 obs. of 5 variables
prec_data	12 obs. of 3 variables
product_data	5 obs. of 2 variables
radar_data	6 obs. of 3 variables
response_data	int [1:3, 1:4] 1 2 0 1 1 1 1 0 1 0 ...
responses	12 obs. of 5 variables
sales_data	3 obs. of 2 variables
sales_matrix	num [1:3, 1:3] 2000 2200 2400 1500 1800 1600 1200 1400 ...
sorted_data	5 obs. of 3 variables

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3. Build a table to show the survey response data for each respondent. Label the table elements..

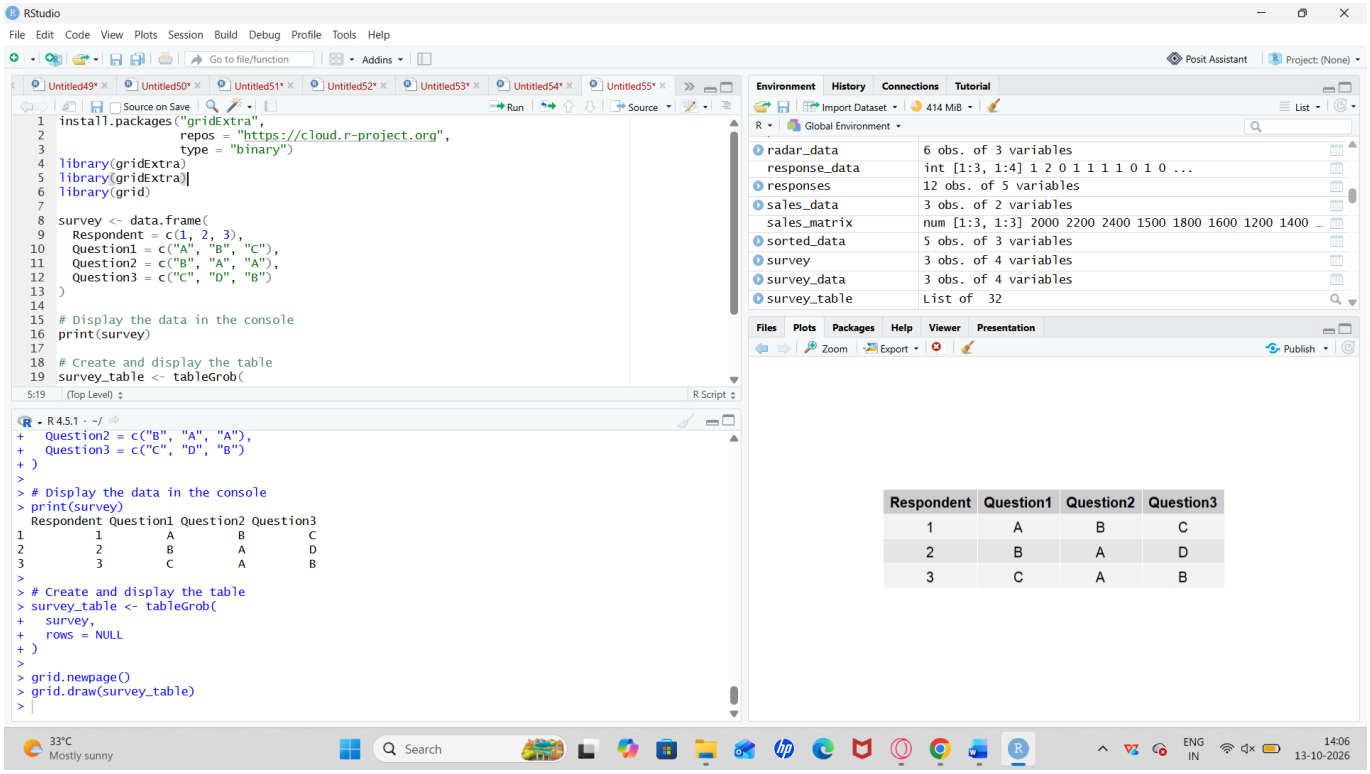
```
install.packages("gridExtra",
                 repos = "https://cloud.r-project.org",
                 type = "binary")
library(gridExtra)
library(gridExtra)
library(grid)

survey <- data.frame(
  Respondent = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)

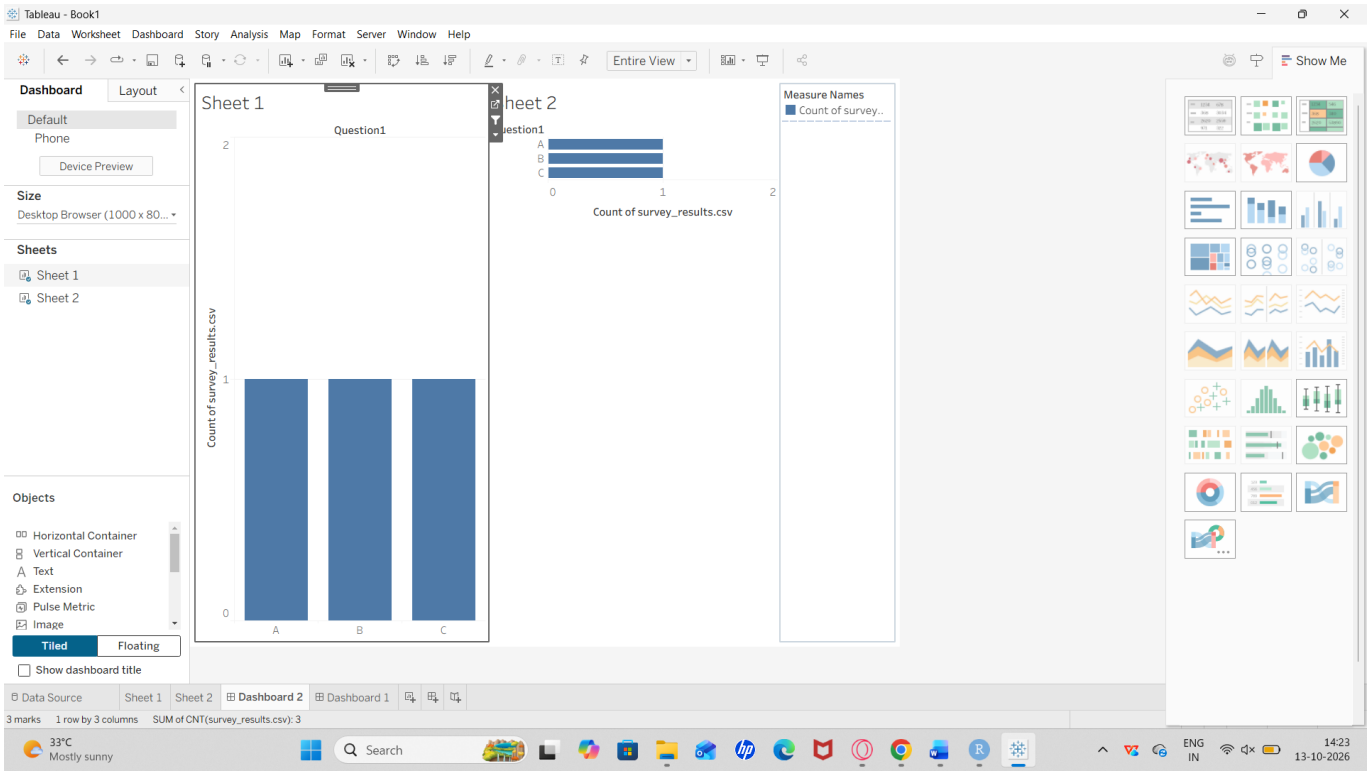
# Display the data in the console
print(survey)

# Create and display the table
survey_table <- tableGrob(
  survey,
  rows = NULL
)

grid.newpage()
grid.draw(survey_table)
```



4.



PROCEDURE

1. Create a survey dataset containing respondent IDs and answers to three questions.
2. Install and load the required R packages, such as ggplot2, fmsb, and gridExtra.
3. Create a stacked bar chart to display the distribution of answers for Question 1.
4. Generate a radar chart to visualize the overall response patterns across all three questions.
5. Create a data table showing the responses of each respondent.
6. Export the survey dataset to a CSV file using the write.csv() function.
7. Import the CSV file into Tableau and create separate worksheets for the visualizations.
8. Combine the worksheets into a single dashboard and add filters for interactive exploration.

Result

Thus, survey data visualization was successfully performed using R programming by creating a stacked bar chart, radar chart, and response table. The dataset was exported to CSV format and prepared for integration into a Tableau dashboard, enabling users to explore and compare survey responses interactively.